

Calculations

System of Units

- Length: Kilometers (km)
- Mass: Solar Masses ($M_{\odot}$)
- Time: Seconds (s)

Relativistic Units

Many of the calculations utilize a relativistic system of units where the speed of light (c) is a dimensionless value equal to 1. This leads to the ability to express length as a unit of time. Further reading.

$$c = L \cdot T^{-1}$$

$$1 = L \cdot T^{-1}$$

$$T = L$$

where L is a unit of length and T is a unit of time. To convert an meters-kilograms-seconds (MKS) value to relativistic units, simply use the following.

$$c = 1 = 2.99792458 \times 10^8 m \cdot s^{-1}$$

$$1s = 2.99792458 \times 10^8 m$$

Time Conversion Factor

Since the unit of length for this project is kilometers, I'm instead using the following conversion factor.

$$1s = 2.99792458 \times 10^8 m$$

$$1s = 2.99792458 \times 10^5 (10^3 m)$$

$$1s = 2.99792458 \times 10^5 km$$

Energy Density Conversion Factor

Though the user interface has energy density and pressure in terms of $MeV \cdot fm^{-3}$, the solver expects different units. To accomodate this, it's necessary to convert to to $M_{\odot} \cdot km^{-3}$.

First, start with energy in terms of SI units.

$$1J = 1kgm^2s^{-2}$$

Find 1 kg in terms of $M_{\odot}$.

$$1M_{\odot} = 1.988416 \times 10^{30} kg$$

$$1kg = 5.02912871 \times 10^{-31} M_{\odot}$$

Find 1 m in terms of km .

$$1km = 10^3 m$$

$$1m = 10^{-3} km$$

Using time conversion factor, plug into the formula for a joule.

$$1J = (5.02912871 \times 10^{-31} M_{\odot}) \cdot (10^{-3} km)^2 \cdot (2.99792458 \times 10^5 km)^{-2}$$

$$1J = 5.59566034 \times 10^{-48} M_{\odot}$$

Given the conversion factor between electronvolts and joules

$$1eV = 1.602176634 \times 10^{-19} J$$

$$1J = 6.241509074 \times 10^{18} eV$$

Therefore

$$6.241509074 \times 10^{18} eV = 5.59566034 \times 10^{-48} M_{\odot}$$

$$1eV = 8.96523625 \times 10^{-67} M_{\odot}$$

$$10^{-6} MeV = 8.96523625 \times 10^{-67} M_{\odot}$$

$$1MeV = 8.96523625 \times 10^{-61} M_{\odot}$$

Next, find km in terms of fm

$$1m = 10^{15} fm = 10^{-3} km$$

$$1fm = 10^{-18} km$$

$$1fm^3 = 10^{-54} km^3$$

Now, combine these two equations

$$1 \frac{MeV}{fm^3} = \frac{8.96523625 \times 10^{-61} M_{\odot}}{10^{-54} km^3}$$

$$1 \frac{MeV}{fm^3} = 8.96523625 \times 10^{-7} \frac{M_{\odot}}{km^3}$$

This gives the conversion factor between $MeV \cdot fm^{-3}$ and $M_{\odot} \cdot km^{-3}$

Gravitational Constant

Starting from non-relativistic units (source):

$$G = 1.3271244002 \times 10^{11} \frac{km^3}{M_{\odot} \cdot s^2}$$

$$G = 1.3271244002 \times 10^{11} \frac{km^3}{M_{\odot} \cdot (2.99792458 \times 10^5 km)^2}$$

$$G = \frac{1.3271244002 \times 10^{11}}{2.99792458 \times 10^{10}} \frac{km^3}{M_{\odot} \cdot km^2}$$

$$G = 4.426810498 \frac{km}{M_{\odot}}$$